

EECS 510 Final Project Report

Part 1: Design

Purpose

The purpose of this language is to multiply two binary numbers together. When a valid string is processed, the result with no extraneous characters (blank spots on either side of the result are fine) should be left on the tape.

Alphabet

Because this is a Turing machine, there will be a language alphabet and an auxiliary alphabet. When a string is processed, the intended behavior is that the input and output will consist exclusively of characters in the language alphabet; all characters outside of the language alphabet may be used for any purpose, though they should be removed before the output is returned.

For the language alphabet, since we are simply multiplying two binary numbers, we will simply have '0', '1', and the asterisk '*' to indicate the operator for multiplication.

For our auxiliary alphabet, we will have four characters to represent numbers that have been read: 'A', 'B', 'X', and 'Y'. The former two will be used to indicate that a number has been read, though we might want to do things with it later, and the latter two indicate that the value should not be changed until we wish to return the value. X and A map to 0, and B and Y map to 1.

In addition, we use two additional characters, 'S' and 'E'. These characters are used to surround the output before reversing its order. When this reversing process reaches E, we know that we have finished reversing the string and simply need to remove S and E before halting.

Semantics

An input accepted by the language should consist of a string of binary numbers with non-zero length, followed by an asterisk, followed by another string of binary numbers. No more than one asterisk is allowed in an input string. In addition, the input should have no whitespace.

Examples of valid input strings:

- 0*0
- 1*1
- 00*00
- 110101*011101

Examples of invalid input strings

- 1101*
- *0
- 111*111 11
- 1 * 1

Part 2: Grammar

When I initially implemented my automata, I originally designed it as a Turing machine. I have converted that Turing machine to a grammar here. I have done my best to convert manually and proofread it and I had Microsoft copilot proofread it as well. To reduce the size of this document, the grammar's rules have been arranged into three columns.

$V_00 \rightarrow 0V_1$	$1F_0X \rightarrow F_01X$	$0F_3X \rightarrow F_30X$
$V_01 \rightarrow 1V_1$	$XF_0X \rightarrow F_0XX$	$1F_3X \rightarrow F_31X$
$V_10 \rightarrow 0V_1$	$\square F_0X \rightarrow F_0\square X$	$XF_3X \rightarrow F_3XX$
$V_11 \rightarrow 1V_1$	$F_00 \rightarrow XF_1$	$YF_3X \rightarrow F_3YX$
$V_1^* \rightarrow ^*V_2$	$F_01 \rightarrow XS_0$	$^*F_3X \rightarrow F_3^*X$
$V_20 \rightarrow 0V_2$	$F_0\square \rightarrow \square C_0$	$\square F_3Y \rightarrow F_3\square Y$
$V_21 \rightarrow 1V_2$	$F_10 \rightarrow 0F_1$	$0F_3Y \rightarrow F_30Y$
$0V_2\square \rightarrow V_30\square$	$F_11 \rightarrow 1F_1$	$1F_3Y \rightarrow F_31Y$
$1V_2\square \rightarrow V_31\square$	$F_1X \rightarrow XF_1$	$XF_3Y \rightarrow F_3XY$
$0V_30 \rightarrow V_400$	$F_1^* \rightarrow ^*F_1$	$YF_3Y \rightarrow F_3YY$
$0V_31 \rightarrow V_401$	$F_1\square \rightarrow \square F_2$	$^*F_3Y \rightarrow F_3^*Y$
$1V_30 \rightarrow V_410$	$F_2X \rightarrow XF_2$	$\square F_30 \rightarrow F_3\square 0$
$1V_31 \rightarrow V_411$	$F_2Y \rightarrow YF_2$	$0F_30 \rightarrow F_300$
$0V_40 \rightarrow V_400$	$XF_2\square \rightarrow F_3XX$	$1F_30 \rightarrow F_310$
$0V_41 \rightarrow V_401$	$YF_2\square \rightarrow F_3YX$	$XF_30 \rightarrow F_3X0$
$1V_40 \rightarrow V_410$	$XF_20 \rightarrow F_3XX$	$YF_30 \rightarrow F_3Y0$
$1V_41 \rightarrow V_411$	$YF_20 \rightarrow F_3YX$	$^*F_30 \rightarrow F_3^*0$
$0V_4^* \rightarrow F_00^*$	$XF_21 \rightarrow F_3XY$	$\square F_31 \rightarrow F_3\square 1$
$1V_4^* \rightarrow F_01^*$	$YF_21 \rightarrow F_3YY$	$0F_31 \rightarrow F_301$
$0F_0X \rightarrow F_00X$	$\square F_3X \rightarrow F_3\square X$	$1F_31 \rightarrow F_311$

$XF_31 \rightarrow F_3X1$	$*S_1B \rightarrow S_1*B$	$BS_{2.1}\square \rightarrow S_4BA$
$YF_31 \rightarrow F_3Y1$	$S_10 \rightarrow AS_2$	$S_3A \rightarrow AS_3$
$*F_31 \rightarrow F_3*1$	$S_11 \rightarrow BS_3$	$S_3B \rightarrow BS_3$
$\square F_3\square \rightarrow F_3\square\square$	$S_1^* \rightarrow *S_{C0}$	$S_3\square \rightarrow \square S_{3.1}$
$0F_3\square \rightarrow F_30\square$	$S_2A \rightarrow AS_2$	$S_{3.1}A \rightarrow AS_{3.1}$
$1F_3\square \rightarrow F_31\square$	$S_2B \rightarrow BS_2$	$S_{3.1}B \rightarrow BS_{3.1}$
$XF_3\square \rightarrow F_3X\square$	$S_2\square \rightarrow \square S_{2.1}$	$S_{3.1}X \rightarrow XS_{3.1}$
$YF_3\square \rightarrow F_3Y\square$	$S_{2.1}A \rightarrow AS_{2.1}$	$S_{3.1}Y \rightarrow YS_{3.1}$
$*F_3\square \rightarrow F_3*\square$	$S_{2.1}B \rightarrow BS_{2.1}$	$\square S_{3.1}0 \rightarrow S_4\square B$
$XF_3^* \rightarrow F_0X^*$	$S_{2.1}X \rightarrow XS_{2.1}$	$S_{3.1}1 \rightarrow AS_{3.2}$
$S_00 \rightarrow 0S_0$	$S_{2.1}Y \rightarrow YS_{2.1}$	$\square S_{3.1}\square \rightarrow S_4\square B$
$S_01 \rightarrow 1S_0$	$\square S_{2.1}0 \rightarrow S_4\square A$	$XS_{3.1}0 \rightarrow S_4XB$
$S_0X \rightarrow XS_0$	$\square S_{2.1}1 \rightarrow S_4\square B$	$XS_{3.1}\square \rightarrow S_4XB$
$S_0^* \rightarrow *S_0$	$\square S_{2.1}\square \rightarrow S_4\square A$	$YS_{3.1}0 \rightarrow S_4YB$
$0S_0\square \rightarrow S_10\square$	$XS_{2.1}0 \rightarrow S_4XA$	$YS_{3.1}\square \rightarrow S_4YB$
$1S_0\square \rightarrow S_11\square$	$XS_{2.1}1 \rightarrow S_4XB$	$AS_{3.1}0 \rightarrow S_4AB$
$AS_1A \rightarrow S_1AA$	$XS_{2.1}\square \rightarrow S_4XA$	$AS_{3.1}\square \rightarrow S_4AB$
$BS_1A \rightarrow S_1BA$	$YS_{2.1}0 \rightarrow S_4YA$	$BS_{3.1}0 \rightarrow S_4BB$
$0S_1A \rightarrow S_10A$	$YS_{2.1}1 \rightarrow S_4YB$	$BS_{3.1}\square \rightarrow S_4BB$
$1S_1A \rightarrow S_11A$	$YS_{2.1}\square \rightarrow S_4YA$	$S_{3.2}1 \rightarrow 0S_{3.2}$
$*S_1A \rightarrow S_1^*A$	$AS_{2.1}0 \rightarrow S_4AA$	$AS_{3.2}0 \rightarrow S_4A1$
$AS_1B \rightarrow S_1AB$	$AS_{2.1}1 \rightarrow S_4AY$	$AS_{3.2}\square \rightarrow S_4A1$
$BS_1B \rightarrow S_1BB$	$AS_{2.1}\square \rightarrow S_4AA$	$0S_{3.2}0 \rightarrow S_401$
$0S_1B \rightarrow S_10B$	$BS_{2.1}0 \rightarrow S_4BA$	$0S_{3.2}\square \rightarrow S_401$
$1S_1B \rightarrow S_11B$	$BS_{2.1}1 \rightarrow S_4BB$	$XS_4X \rightarrow S_4XX$

$$XS_4Y \rightarrow S_4XY$$

$$XS_4A \rightarrow S_4XA$$

$$XS_4B \rightarrow S_4XB$$

$$XS_40 \rightarrow S_4X0$$

$$XS_41 \rightarrow S_4X1$$

$$YS_4X \rightarrow S_4YX$$

$$YS_4Y \rightarrow S_4YY$$

$$YS_4A \rightarrow S_4YA$$

$$YS_4B \rightarrow S_4YB$$

$$YS_40 \rightarrow S_4Y0$$

$$YS_41 \rightarrow S_4Y1$$

$$AS_4X \rightarrow S_4AX$$

$$AS_4Y \rightarrow S_4AY$$

$$AS_4A \rightarrow S_4AA$$

$$AS_4B \rightarrow S_4AB$$

$$AS_40 \rightarrow S_4A0$$

$$AS_41 \rightarrow S_4A1$$

$$BS_4X \rightarrow S_4BX$$

$$BS_4Y \rightarrow S_4BY$$

$$BS_4A \rightarrow S_4BA$$

$$BS_4B \rightarrow S_4BB$$

$$BS_40 \rightarrow S_4B0$$

$$BS_41 \rightarrow S_4B1$$

$$0S_4X \rightarrow S_40X$$

$$0S_4Y \rightarrow S_40Y$$

$$0S_4A \rightarrow S_40A$$

$$0S_4B \rightarrow S_40B$$

$$0S_40 \rightarrow S_400$$

$$0S_41 \rightarrow S_401$$

$$1S_4X \rightarrow S_41X$$

$$1S_4Y \rightarrow S_41Y$$

$$1S_4A \rightarrow S_41A$$

$$1S_4B \rightarrow S_41B$$

$$1S_40 \rightarrow S_410$$

$$1S_41 \rightarrow S_411$$

$$\square S_4X \rightarrow S_4\square X$$

$$\square S_4Y \rightarrow S_4\square Y$$

$$\square S_4A \rightarrow S_4\square A$$

$$\square S_4B \rightarrow S_4\square B$$

$$\square S_40 \rightarrow S_4\square 0$$

$$\square S_41 \rightarrow S_4\square 1$$

$$AS_4\square \rightarrow S_1A\square$$

$$BS_4\square \rightarrow S_1B\square$$

$$S_{C0}A \rightarrow 0S_{C0}$$

$$S_{C0}B \rightarrow 1S_{C0}$$

$$S_{C0}\square \rightarrow \square S_{C1}$$

$$S_{C1}X \rightarrow XS_{C1}$$

$$S_{C1}Y \rightarrow YS_{C1}$$

$$S_{C1}A \rightarrow XS_{C2}$$

$$S_{C1}B \rightarrow YS_{C2}$$

$$S_{C2}0 \rightarrow 0S_{C2}$$

$$S_{C2}1 \rightarrow 1S_{C2}$$

$$S_{C2}A \rightarrow 0S_{C2}$$

$$S_{C2}B \rightarrow 1S_{C2}$$

$$0S_{C2}\square \rightarrow S_{C3}0\square$$

$$1S_{C2}\square \rightarrow S_{C3}1\square$$

$$AS_{C2}\square \rightarrow S_{C3}A\square$$

$$BS_{C2}\square \rightarrow S_{C3}B\square$$

$$0S_{C3}0 \rightarrow S_{C3}00$$

$$0S_{C3}1 \rightarrow S_{C3}01$$

$$0S_{C3}X \rightarrow S_{C3}0X$$

$$0S_{C3}Y \rightarrow S_{C3}0Y$$

$$0S_{C3}\square \rightarrow S_{C3}0\square$$

$$1S_{C3}0 \rightarrow S_{C3}10$$

$$1S_{C3}1 \rightarrow S_{C3}11$$

$$1S_{C3}X \rightarrow S_{C3}1X$$

$$1S_{C3}Y \rightarrow S_{C3}1Y$$

$$1S_{C3}\square \rightarrow S_{C3}1\square$$

$$XS_{C3}0 \rightarrow S_{C3}X0$$

$$XS_{C3}1 \rightarrow S_{C3}X1$$

$$XS_{C3}X \rightarrow S_{C3}XX$$

$$XS_{C3}Y \rightarrow S_{C3}XY$$

$$XS_{C3}\square \rightarrow S_{C3}X\square$$

$$YS_{C3}0 \rightarrow S_{C3}Y0$$

$$YS_{C3}1 \rightarrow S_{C3}Y1$$

$$YS_{C3}X \rightarrow S_{C3}YX$$

$$YS_{C_3}Y \rightarrow S_{C_3}YY$$

$$YS_{C_3}\square \rightarrow S_{C_3}Y\square$$

$$\square S_{C_3}0 \rightarrow S_{C_3}\square 0$$

$$\square S_{C_3}1 \rightarrow S_{C_3}\square 1$$

$$\square S_{C_3}X \rightarrow S_{C_3}\square X$$

$$\square S_{C_3}Y \rightarrow S_{C_3}\square Y$$

$$*S_{C_3}0 \rightarrow S_{C_3}*0$$

$$*S_{C_3}1 \rightarrow S_{C_3}*1$$

$$XS_{C_3}^* \rightarrow F_0X^*$$

$$C_00 \rightarrow \square C_0$$

$$C_01 \rightarrow \square C_0$$

$$C_0^* \rightarrow \square C_0$$

$$C_0X \rightarrow \square C_0$$

$$C_0\square \rightarrow SC_1$$

$$C_10 \rightarrow 0C_1$$

$$C_11 \rightarrow 1C_1$$

$$C_1X \rightarrow XC_1$$

$$C_1Y \rightarrow YC_1$$

$$0C_1\square \rightarrow C_20E$$

$$1C_1\square \rightarrow C_21E$$

$$XC_1\square \rightarrow C_2XE$$

$$YC_1\square \rightarrow C_2YE$$

$$0C_20 \rightarrow C_200$$

$$1C_20 \rightarrow C_210$$

$$XC_20 \rightarrow C_2X0$$

$$YC_20 \rightarrow C_2Y0$$

$$SC_20 \rightarrow C_2S0$$

$$0C_21 \rightarrow C_201$$

$$1C_21 \rightarrow C_211$$

$$XC_21 \rightarrow C_2X1$$

$$YC_21 \rightarrow C_2Y1$$

$$SC_21 \rightarrow C_2S1$$

$$0C_2X \rightarrow C_200$$

$$1C_2X \rightarrow C_210$$

$$XC_2X \rightarrow C_2X0$$

$$YC_2X \rightarrow C_2Y0$$

$$SC_2X \rightarrow C_2S0$$

$$0C_2Y \rightarrow C_201$$

$$1C_2Y \rightarrow C_211$$

$$XC_2Y \rightarrow C_2X1$$

$$YC_2Y \rightarrow C_2Y1$$

$$SC_2Y \rightarrow C_2S1$$

$$C_2S \rightarrow SC_3$$

$$C_3\square \rightarrow \square C_3$$

$$SC_30 \rightarrow C_4S\square$$

$$\square C_30 \rightarrow C_4\square\square$$

$$SC_31 \rightarrow C_5S\square$$

$$\square C_31 \rightarrow C_5\square\square$$

$$\square C_3E \rightarrow C_6\square\square$$

$$\square C_4\square \rightarrow C_4\square\square$$

$$SC_4\square \rightarrow C_4S\square$$

$$\square C_4S \rightarrow C_{4.1}\square S$$

$$0C_4S \rightarrow C_{4.1}0S$$

$$1C_4S \rightarrow C_{4.1}1S$$

$$0C_{4.1}0 \rightarrow C_{4.1}00$$

$$1C_{4.1}0 \rightarrow C_{4.1}10$$

$$\square C_{4.1}0 \rightarrow C_{4.1}\square 0$$

$$0C_{4.1}1 \rightarrow C_{4.1}01$$

$$1C_{4.1}0 \rightarrow C_{4.1}11$$

$$\square C_{4.1}0 \rightarrow C_{4.1}\square 1$$

$$C_{4.1}\square \rightarrow 0C_{4.2}$$

$$C_{4.2}0 \rightarrow 0C_{4.2}$$

$$C_{4.2}1 \rightarrow 0C_{4.2}$$

$$C_{4.2}S \rightarrow SC_3$$

$$\square C_5\square \rightarrow C_5\square\square$$

$$SC_5\square \rightarrow C_5S\square$$

$$\square C_5S \rightarrow C_{5.1}\square S$$

$$0C_5S \rightarrow C_{5.1}0S$$

$$1C_5S \rightarrow C_{5.1}1S$$

$$0C_{5.1}0 \rightarrow C_{5.1}00$$

$$1C_{5.1}0 \rightarrow C_{5.1}10$$

$$\square C_{5.1}0 \rightarrow C_{5.1}\square 0$$

$$0C_{5.1}1 \rightarrow C_{5.1}01$$

$$1C_{5.1}0 \rightarrow C_{5.1}10$$

$$\square C_{5.1}0 \rightarrow C_{5.1}\square 0$$

$$C_{5.1}\square \rightarrow 1C_{5.2}$$

$$C_{5.2}0 \rightarrow 0C_{5.2}$$

$$C_{5.2}1 \rightarrow 0C_{4.2}$$

$$C_{5.2}S \rightarrow SC_3$$

$$\square C_6\square \rightarrow C_6\square\square$$

$$SC_6\square \rightarrow C_6S\square$$

$$0C_6S \rightarrow H0\square$$

$$1C_6S \rightarrow H1\square$$

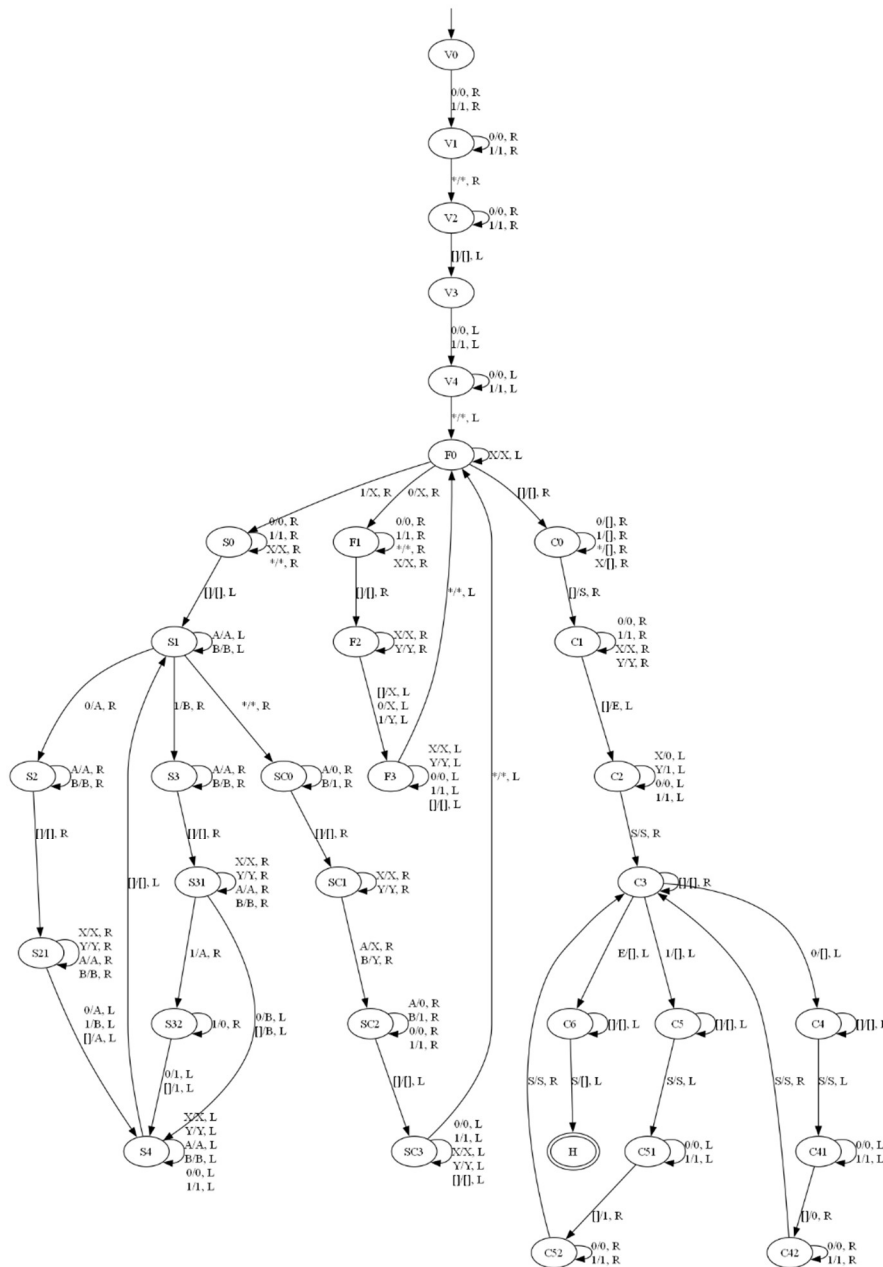
$$\square H \rightarrow H$$

$$H\square \rightarrow H$$

$$H \rightarrow \lambda$$

Part 3: Automaton

This is a graph I generated using GraphViz. I'm not particularly experienced with any graph creation programs, so it's not perfect. That being said, I think it's at least serviceable. The graph and the gv file used to create it can be found on GitHub.



Part 4: Code Notes

The code can be run with python3 on main.py. It requires two additional arguments: a path to the automata, and a file containing test input strings. In addition, a flag “-human” can be appended to the end to show the output of the string and make the results more readable (as opposed to simply “accept” and “reject”).

When I made and tested my code, I used Python version 3.12.3.

Because recursively enumerable languages have a chance to not halt, there is a limit to the number of transitions one can take. This limit is a function of the length of the input string: $f(l) = 16l^2 + 16$

For your convenience, this is the prompt that I used:

```
python3 main.py ./TestInputs/BinaryMultiplier.aut  
./TestInputs/BinaryInputs.txt
```

```
python3 main.py ./TestInputs/BinaryMultiplier.aut ./TestInputs/BinaryInputs.txt
```

File Formatting

Regular and recursively enumerable languages are supported by my code. To signify that a language is regular, the first line of the input file should be “REG”. If the machine is recursively enumerable, the first line should be “TUR”.

Comments are supported throughout the file, except for the first line, which must always be the machine type. To use a comment, simply make the first character in the line a pound sign ‘#’.

If a transition is not defined and the machine would need to use that transition, the machine will simply reject the input.

Regular Formatting

Line Number	Contents of line
1	Type of machine “REG”
2	Language alphabet
3	List of states
4	List of <i>accepting</i> states. All states listed here must appear in line 3 as well.
5 and onwards	All transitions, with the following format: [FROM] [CONSUMED] [TO]

Turing-Specific Formatting

Line Number	Contents of line
1	Type of machine “REG”
2	Language alphabet
3	List of states
4	List of <i>accepting</i> states. All states listed here must appear in line 3 as well.
5 and onwards	All transitions, with the following format: [FROM] [CONSUMED_CHAR] [REPLACED] [DIRECTION] [TO] For the purpose of the consumed and replaced characters, an empty string represents a blank, and ‘L’ and ‘R’ represent directions.

Input Formatting

The file containing the input strings is simply a list of strings, separated by newline. Comments are not supported.